In [1]:
```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import numpy as np
import cufflinks as cf
from plotly.offline import iplot
import warnings
warnings.filterwarnings('ignore')
```

pd.read_csv('D:\IPL.csv')

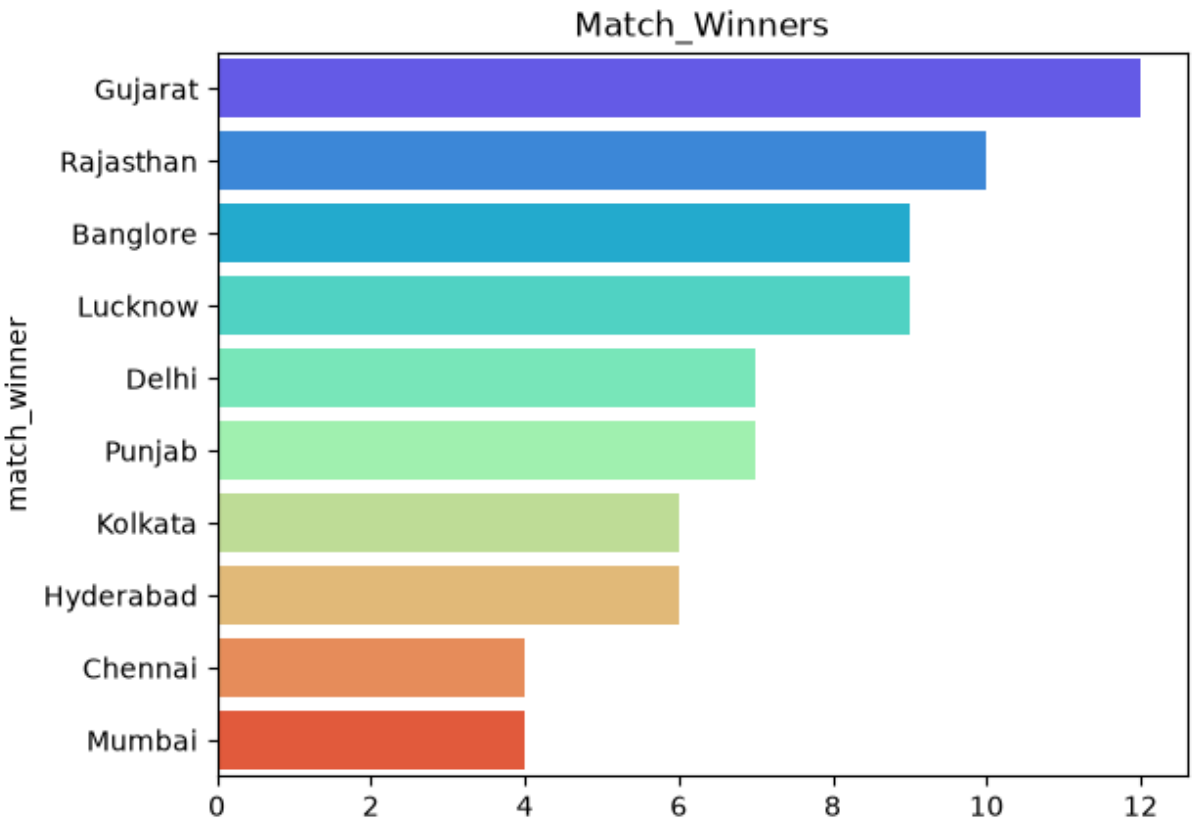
In [2]:
```
df=pd.read_csv('D:\IPL.csv')
```

In [ ]:
```
# IPL PROJECT USING SEABORN AND MATPLOTLIB
```

In [ ]:
```
1.Which team won the most matches?
```

In [46]:
```
match_winners=df['match_winner'].value_counts()
sns.barplot(x=match_winners.values,y=match_winners.index,palette='rainbow')
plt.title('Match_Winners')
```

Out[46]:
```
Text(0.5, 1.0, 'Match_Winners')
```

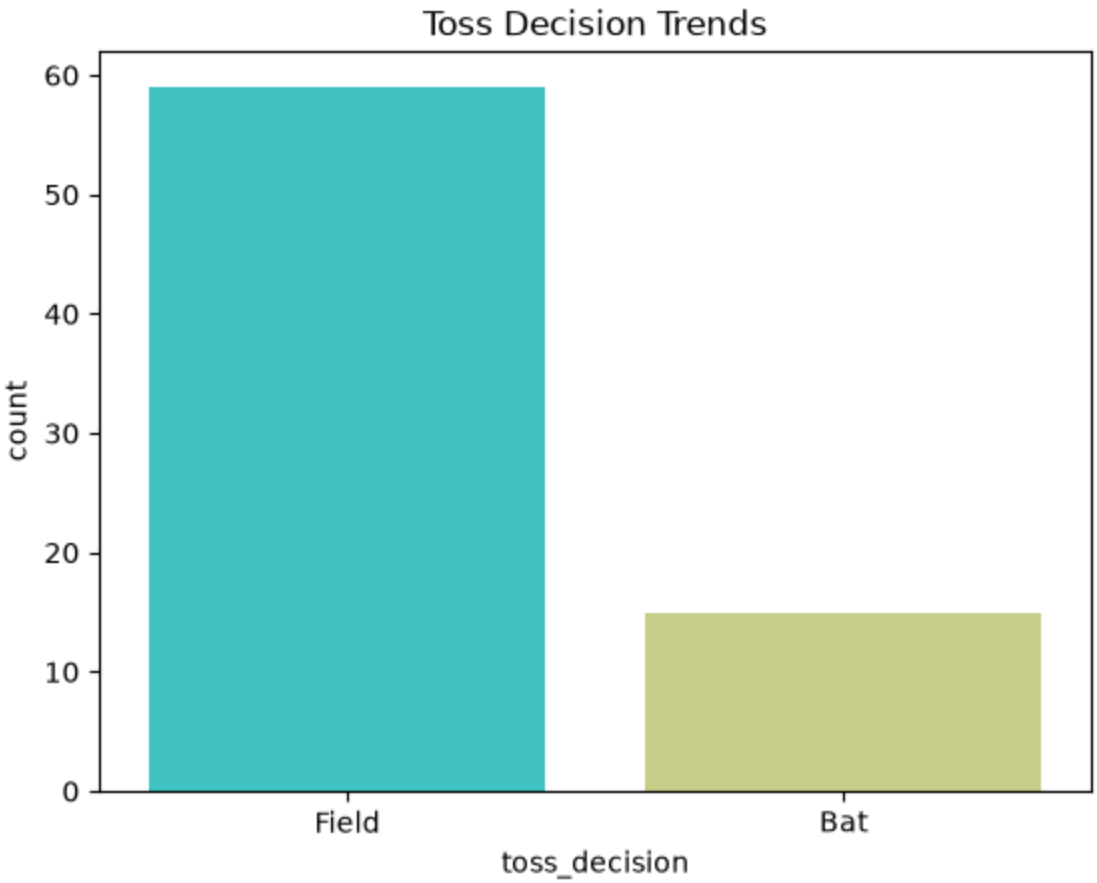


In [ ]:
```
2.Toss Decision Trends
```

In [53]:
```
df['toss_decision']
sns.countplot(x=df['toss_decision'],palette='rainbow')
```

```
plt.title('Toss Decision Trends')
```

Out[53]: Text(0.5, 1.0, 'Toss Decision Trends')



In [ ]:
```
3.Toss_Winner Vs Match_Winner
```

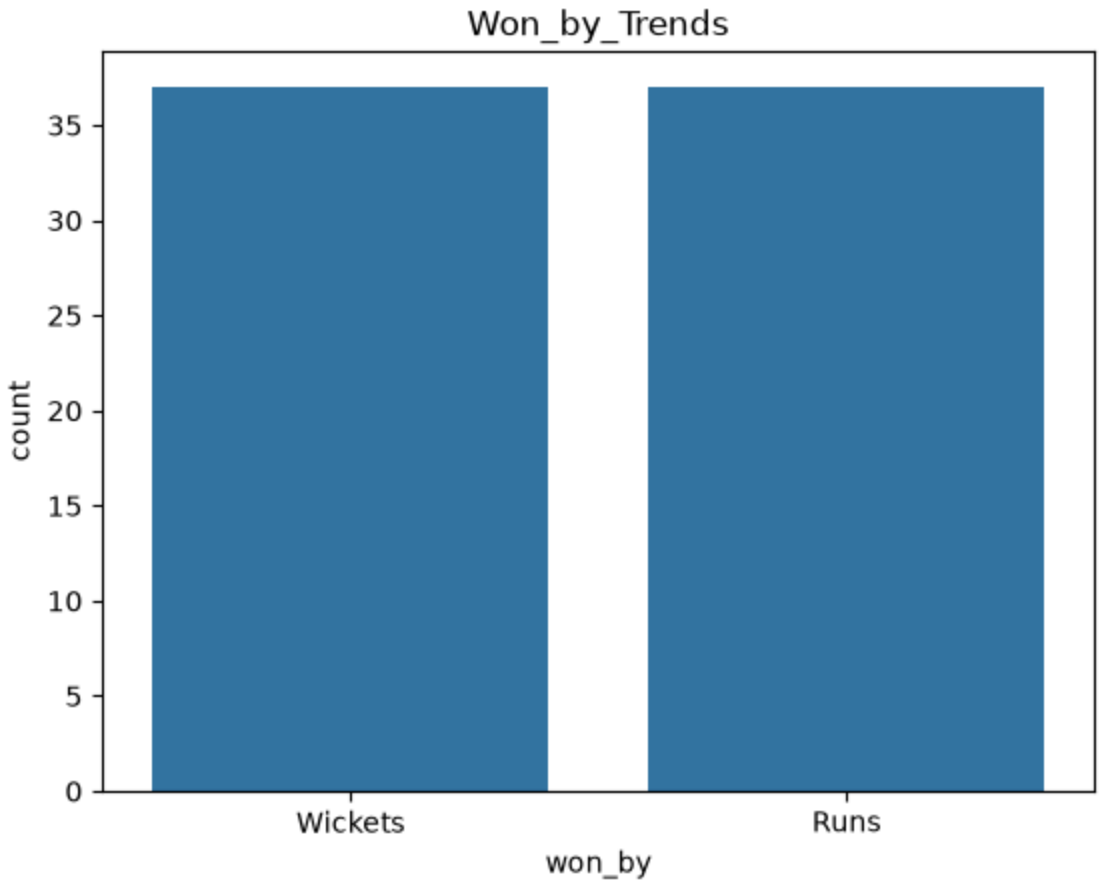
In [63]:
```
count=df[df['toss_winner']==df['match_winner']]['match_id'].count()
percentage=(count/df.shape[0])*100
percentage.round(2)
```

Out[63]: np.float64(48.65)

In [ ]:
```
4.How do team wins(Runs vs Wickets)
```

In [70]:
```
df[ 'won_by']
sns.countplot( x=df['won_by'])
plt.title('Won_by_Trends')
```

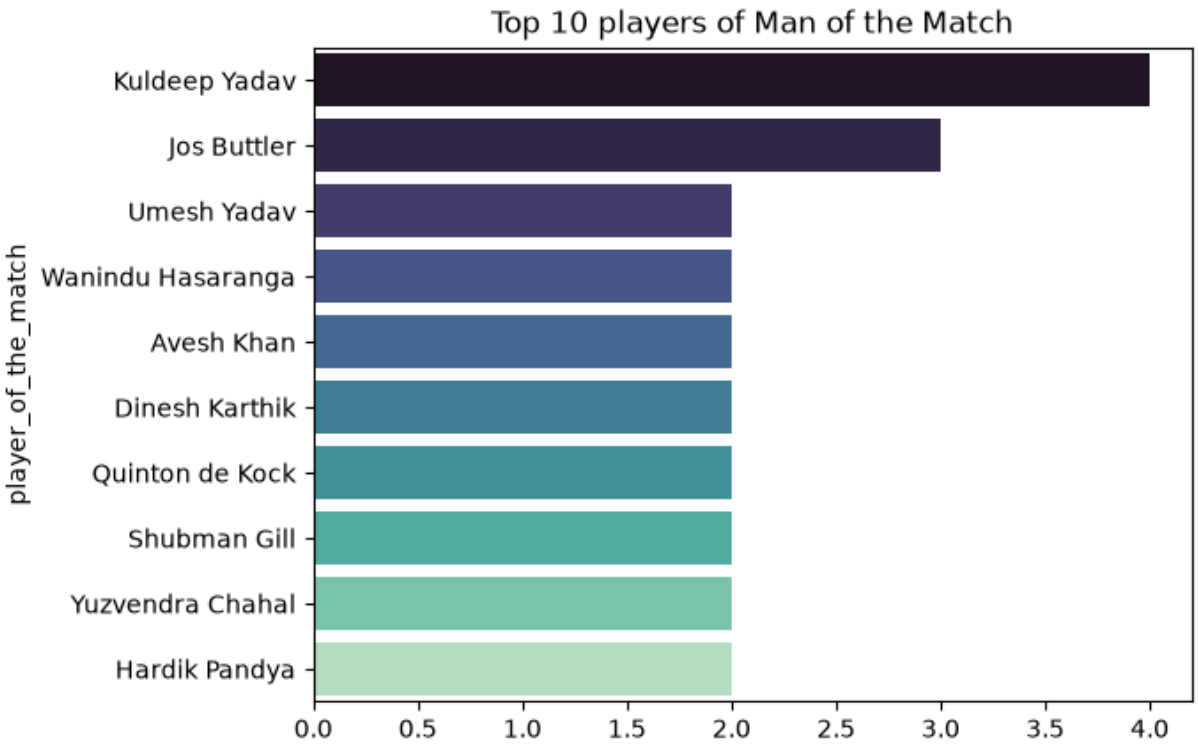
Out[70]: Text(0.5, 1.0, 'Won_by_Trends')



In [ ]:
```
5.Key Player Performance
 1.Player of the Match
```

In [83]:
```
count=df['player_of_the_match'].value_counts().head(10)
sns.barplot(x=count.values,y=count.index,palette='mako')
plt.title('Top 10 players of Man of the Match')
```

Out[83]:
```
Text(0.5, 1.0, 'Top 10 players of Man of the Match')
```

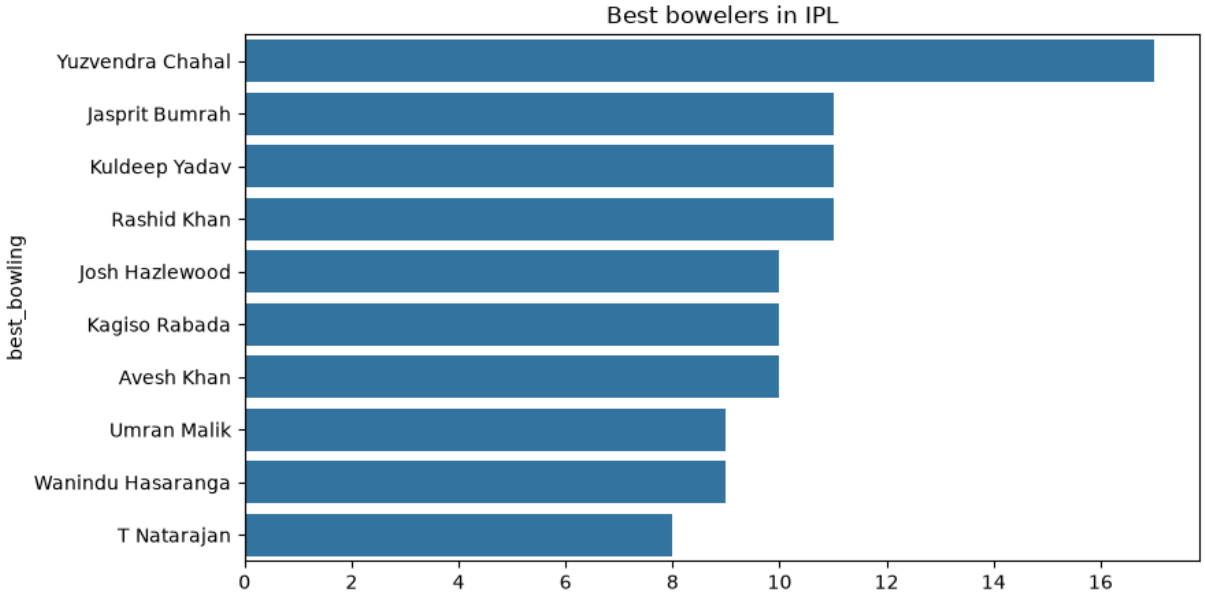


In [ ]:
```
2. Best Bowling Figure
```

In [103…
```
# df['best_bowling_figure'].
df['highest_wicket']=df['best_bowling_figure'].apply(lambda x:x.split('--')[0])
df['highest_wicket']=df['highest_wicket'].astype(int)
df['highest_wicket']

best_bowlers=df.groupby('best_bowling')['highest_wicket'].sum().sort_values(ascendi
figure=plt.figure(figsize=(9,5))
sns.barplot(x=best_bowlers.values,y=best_bowlers.index)
plt.title('Best bowelers in IPL')
```

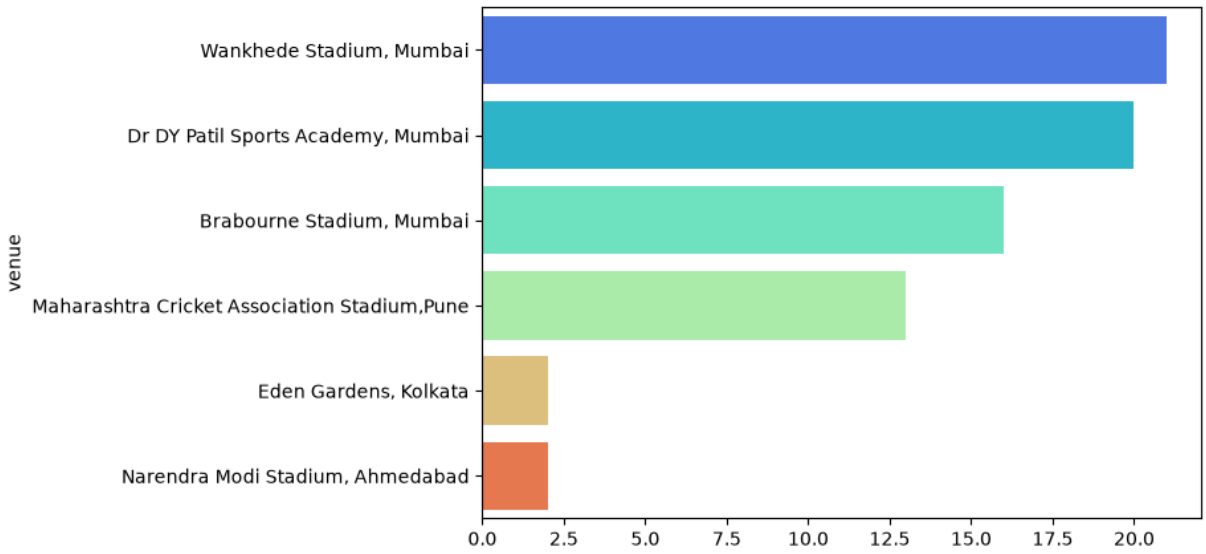
Out[103…
```
Text(0.5, 1.0, 'Best bowelers in IPL')
```



In [104…
```
# VENUE ANALYSIS
```

In [113…
```
# MOST MATCHES PLAYED BY VENUE
venue=df['venue'].value_counts()
figure=plt.figure(figsize=(7,5))
sns.barplot(x=venue.values,y=venue.index,palette='rainbow')
```

Out[113… <Axes: ylabel='venue'>



In [124…
```
# Custom Questions and Insights
# 1 Who won the highest margin by runs

df[df['won_by']=='Runs'].sort_values(by='margin',ascending=False).head(1)[['match_w
```

Out[124…

<style scoped> .dataframe tbody tr th:only-of-type { vertical-align: middle; } .dataframe tbody tr th { vertical-align: top; } .dataframe thead th { text-align: right; } </style> <thead> <th> </th> <th>match_winner</th> <th>won_by</th> <th>margin</th> </thead> <tbody> <th>54</th> </tbody> <tbody></tbody>

Chennai Runs 91

In [130…

```
#2 Which player had the highest individual score
df[df['highscore']==df['highscore'].max()][['top_scorer','highscore']]
```

Out[130…

<style scoped> .dataframe tbody tr th:only-of-type { vertical-align: middle; } .dataframe tbody tr th { vertical-align: top; } .dataframe thead th { text-align: right; } </style> <thead> <th> </th> <th>top_scorer</th> <th>highscore</th> </thead> <tbody> <th>65</th> </tbody> <tbody></tbody>

Quinton de Kock 140

In [133…

```
# 3 Which bowler had the highest bowling figure
df[df['best_bowling_figure']==df['best_bowling_figure'].max()][['best_bowling','bes
```

Out[133…

<style scoped> .dataframe tbody tr th:only-of-type { vertical-align: middle; } .dataframe tbody tr th { vertical-align: top; } .dataframe thead th { text-align: right; } </style> <thead> <th> </th> <th>best_bowling</th> <th>best_bowling_figure</th> </thead> <tbody> <th>29</th> </tbody> <tbody></tbody>

Yuzvendra Chahal 5--40